Project 1 Report

Name: Course: CECS 327 **Date:** February 2026
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Overview

In this project, I built a simple UDP client-server application using Python sockets and containerized it using Docker and Docker Compose. The goal was to demonstrate network communication in a distributed system.

What We Did

We wrote a UDP server (`server.py`) that binds to port 3000 and continuously listens for incoming messages, printing each message and the address it came from. We then wrote a UDP client (`client.py`) that connects to the server by its Docker service name and sends a series of messages with a short delay between each one.

Both programs were packaged into separate Docker images using individual Dockerfiles. A `docker-compose.yml` file was used to define both services on the same network, allowing the client to reach the server by hostname. We got stuck on the output not being displayed, but we used -u flag and PYTHONUNBUFFERED to force output to the terminal.

To simulate multiple clients connecting simultaneously, I used Docker Compose's `--scale` flag to spin up 3 client instances at once.

Run command:

docker compose up --build --scale client=3

```
[+] up 2/2
 ✔ Image project-1-server Built
 ✔ Image project-1-client Built
Attaching to client-1, client-2, client-3, server-1
server-1  | listening on 0.0.0.0:3000
server-1  | received from ('172.22.0.3', 52818): hello from client
server-1  | received from ('172.22.0.4', 41685): hello from client
server-1  | received from ('172.22.0.5', 51212): hello from client
server-1  | received from ('172.22.0.3', 52818): hello from client
server-1  | received from ('172.22.0.4', 41685): hello from client
server-1  | received from ('172.22.0.5', 51212): hello from client
server-1  | received from ('172.22.0.3', 52818): hello from client
server-1  | received from ('172.22.0.4', 41685): hello from client
server-1  | received from ('172.22.0.5', 51212): hello from client
client-3 exited with code 0
client-1 exited with code 0
client-2 exited with code 0
```